eLife

Reviewed Preprint
v1 • November 28, 2025
Not revised

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Competing interests: No competing interests declared

Reviewing editor: Bavesh D Kana, University of the Witwatersrand, South Africa

Dark matter of an orchid: metagenome of the microbiome associated with the rhizosphere of *Dactylorhiza traunsteineri*

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eLife Assessment

This study presents a **useful** overview of the taxonomic composition of the microbiome associated with Dactylorhiza traunsteineri, a widely distributed orchid species in Central Europe. The evidence supporting the claims of the authors is **incomplete**, especially when it comes to the (secondary) metabolic pathways found in the metagenome assembled genomes, and requires more substantial analysis to be able to claim that these pathways play a key role in microbiome-orchid symbiosis.

https://doi.org/10.7554/eLife.108126.1.sa3

Abstract

Plant microbiota forms complex co-associations with its host, promoting health in natural environments. Root-associated bacteria (RAB) colonize root compartments and can modulate plant functions by producing bioactive compounds as part of their secondary metabolism. We present an in-depth analysis of the rhizosphere-associated microbiome of the endangered marsh orchid *Dactylorhiza traunsteineri*. Using deep sequencing of 16S rRNA genes, we identified *Proteobacteria*, *Actinobacteria*, *Myxococcota*, *Bacteroidota*, and *Acidobacteria* as predominant phyla associated with the rhizosphere of *D. traunsteineri*. Using deep shotgun metagenomics and de novo assembly, we extracted high-quality metagenome-assembled genomes (MAGs), revealing significant metabolic and biosynthetic potentials of the *D. traunsteineri*'s RABs. Our study offers a comprehensive investigation into the microbial community of the *D. traunsteineri* rhizosphere, highlighting a potential novel source of critical bioactive substances. Our study offers novel insights and a robust platform for future investigations into *D. traunsteineri*'s rhizosphere, crucial for understanding plant-microbe interactions and aiding conservation efforts for endangered orchids.

Introduction

Every large multicellular organism studied so far, including plants, interacts with a plethora of microorganisms in its environment. The plant microbiota, which includes bacteria, fungi, protists, nematodes, and viruses, can form complex co-associations with its host. Plant microbial communities have been shown to promote plant growth, nutrient uptake, as well as pathogen resistance, thus promoting plant health in its natural environment (Richardson and Simpson, 2011, Pieterse et al., 2014, Trivedi et al., 2016, Backer et al., 2018, Gouda et al., 2018, Trivedi et al., 2020). However, our understanding of plant-microbial alliances' functional mechanisms and broad ecological implications is still in its early stages (Martin et al., 2017).

Among the bacterial communities interacting with plants, a subset known as the root-associated bacteria (RAB) colonizes root compartments, such as the rhizosphere (i.e. the narrow zone surrounding and influenced by plant roots growth and metabolic activity) and intraradical regions and has the ability to modulate plant functions (Mendes et al., 2013, Qu et al., 2020, Dobbelaere et al., 2003, Walker et al., 2003). It has been suggested that the holobiont, defined as the host and its associated microbiome, forms a discrete symbiotic ecological unit that has co-evolved to maintain host function and fitness (Hassani et al., 2018, Uroz et al., 2019, Vandenkoornhuyse et al., 2015, Trivedi et al., 2020, Liu et al., 2019). Plants selectively recruit RAB depending on different factors such as host genotype, metabolic profiles, root exudates, and soil physicochemical properties (Bulgarelli et al., 2013, Hassani et al., 2018, Liu et al., 2019, Uroz et al., 2019). Although the plant and its exudates mostly influence this microenvironment, environmental influences also significantly shape its composition (Vives-Peris et al., 2020, Najera et al., 2020). Furthermore, diverse endophytic bacteria possess the ability to produce bioactive compounds of usually low molecular mass, with a wide range of reported functions, as part of their secondary metabolism. These compounds, which are not crucial for the fundamental cellular processes of bacteria in controlled laboratory conditions, often confer selective advantages within natural ecosystems. Such secondary metabolites (SMs) include antibiotics, pigments, toxins, enzyme inhibitors, immunomodulators, effectors of ecological competition or symbiosis, as well as compounds with hormonal activity or particular effects on lipids or carbohydrate metabolism (reviewed in (Fouillaud and Dufosse, 2022)). Many RABs have been reported to enhance plant growth through processes such as nutrient acquisition, hormonal regulation, protection against pathogens, and mitigation of abiotic stresses (reviewed and discussed in (Narayanan and Glick, 2022, Glick, 1995, Olanrewaju et al., 2017, Glick, 2012, Hassani et al., 2018, Mendes et al., 2013, Qu et al., 2020, Trivedi et al., 2020, Uroz et al., 2019, Vandenkoornhuyse et al., 2015)). These RAB, conferring growth benefits to plant hosts, are collectively referred to as plant growth-promoting rhizobacteria (PGPR), which typically belong to the *Proteobacteria*, *Actinobacteria*, *Firmicutes*, and *Bacteroidetes* phyla (Santoyo et al., 2016). Within these phyla, genera such as *Rhizobium*, *Bacillus*, *Pseudomonas*, and *Burkholderia* are prominently recognized as PGPR and have demonstrated functional contributions in mediating seed germination, plant growth and biocontrol of plant disease, notably by conveying secondary metabolites inclusive of antibiotics, volatile compounds, phytohormones and siderophores (Glick, 2012, Kang et al., 2012, Walia et al., 2013, Bloemberg and Lugtenberg, 2001). Notably, the publication of whole-genome sequences of rhizosphere-inhabiting bacteria facilitates the identification of genes involved in the regulation and production of SMs by comparative and functional genomics. However, despite the established knowledge that symbiotic associations between roots and PGPR provide multiple growth benefits to host plants, little is known regarding the role of PGPR and other RAB in shaping plant ecology through their secondary metabolism (Siefert et al., 2019, Kaur and Sharma, 2021).

Orchids, which account for approximately 10% of all seed plants, exhibit intricate associations with diverse microorganisms in their natural habitat (Givnish et al., 2016, Givnish et al., 2015). One example is mycoheterotrophy, an alternative nutritional strategy observed in certain plant species, particularly orchids, which derive sugars and essential nutrients from symbiotic associations with mycorrhizal fungi (Timilsena et al., 2023). The marsh orchid *Dactylorhiza traunsteineri* is an allotetraploid ($2n = 80$), mainly found in the eastern hemisphere (Paun et al., 2010). *Dactylorhiza traunsteineri* is part of the *D. majalis* complex, which has evolved polytopically from unidirectional hybridization between the diploid species *Dactylorhiza fuchsii* and *D. incarnata* (Pillon et al., 2007, Brandrud et al., 2020). It is slender, usually low-grown with narrow leaves and a lax few-flowered inflorescence, has narrow soil moisture and pH tolerance, and grows in calcareous fens, typically on naturally open sites associated with seepage zones (Kull and Hutchings, 2006, Soó, 1980, Baumann and Künkele, 1988, Delforge, 2001). The species occurs in low-nutrient soils, characterized in particular by very low levels of available nitrate and other macro and micronutrients (Wolfe et al., 2023). A recent study documented, at two different localities, a high diversity of fungi associated with the roots of *D. traunsteineri* in its natural environment, in comparison to its sibling *D. majalis* (Emelianova et al., 2024). Although hybrids and allopolyploids are known to be more resistant to environmental changes due to their larger genomic potential, *D. traunsteineri* is classified as endangered by several organizations in Europe and has proven challenging to grow and maintain under artificial environmental conditions, which is a requirement for ex-situ conservation efforts (Paun et al., 2007, Blinova and Uotila, 2012, Bartók et al., 2018, Paun et al., 2011).

This study presents a comprehensive analysis of the rhizosphere-associated bacterial microbiome of *Dactylorhiza traunsteineri*. Our community analysis, based on 16S ribosomal DNA (rDNA)-targeted sequencing, identified *Proteobacteria*, *Actinobacteria*, *Myxococcota*, *Bacteroidota*, and *Acidobacteria* as the predominant phyla within the *D. traunsteineri* rhizosphere. Leveraging deep shotgun metagenomics and de novo metagenome assembly, we extracted 47 MAGs, unveiling significant metabolic and biosynthetic capabilities of the orchid's RAB. Notably, this study represents the first comprehensive investigation into the microbial community of the *D. traunsteineri* rhizosphere, highlighting a potential novel source of critical bioactive substances. Our findings illuminate previously uncharacterized PGPR SM pathways and the discovery of "dark matter" biosynthetic gene clusters, underscoring the rhizosphere's untapped potential for biotechnological and ecological applications. This research advances our understanding of *D. traunsteineri*'s microbial interactions and sets the stage for future studies to exploit these microbial consortia for conservation and bioprospecting efforts.

Materials & methods

Sampling and sample processing

Dactylorhiza traunsteineri individuals (from the *turfosa* subspecies) were isolated from the Kitzbühel alpine town region, in the western Austrian province of Tyrol on the 25th of May 2018. The average temperatures recorded in this region during summer are between 13°C and 26°C, and between 0°C and 6°C during winter, with an annual total rainfall estimated to 1,443 mm, the majority occurring primarily during summer in June (average 188 mm). The individuals were potted and maintained at the Botanical Garden of the University of Vienna under conditions mimicking those of the sampling region. The sampling of the rhizosphere soil was performed on 03.02.2020 under sterile laboratory conditions. The plantlets were carefully uprooted, the soil discarded by shaking the roots, and the adherent soil directly in contact with the root system collected. Additionally, the roots of the plantlets were washed for 5 min with sterile DNA-free laboratory-grade Sartorius avium® filtered water. Samples were collected in sterile 50 ml Cellstar tubes. Both samples were shock-frozen in liquid nitrogen and kept at -80°C until the metagenomic DNA extraction was performed.

Metagenomic DNA extraction

Isolation of the metagenomic DNA was performed from the rhizosphere soil and the plantlet wash lyophilized samples in a sterile environment. The root wash samples were lyophilized using the FreeZone Freeze Dryer system [LabConco; Cat. 700201000]. 500 mg of lyophilized mass from each sample was transferred to a 2 mL screw cap reaction tube with sterile glass beads. Samples were first disrupted using the FastPrep-24 Classic homogenizer [MPBio; Cat. 116004500] at setting level 6 for 30 seconds. 1 mL of 2xCTAB buffer (pH 8.0) [100 mM Tris (pH 8.0); 20 mM EDTA (pH 8.0); 1.4 M NaCl; 2% (w/v) CTAB] with 4 mL β-mercaptoethanol, was preheated at 55°C and added to the samples, before disrupting the biomass a second time using the FastPrep-24 Classic homogenizer, twice for 30 seconds at setting level 5. The samples were then incubated for 20 minutes at 65°C and centrifuged for 1 minute at 10,000 rpm before transferring the supernatant and foam to fresh, sterilized 2 mL reaction tubes. Next, 400 µL Phenol and 400 µL SEVAG [4% (v/v) isoamyl alcohol; 96% (v/v) chloroform] were added to the samples before vigorous mixing and incubation for 1 hour at room temperature. The aqueous phases were transferred to fresh, sterilized 2 mL reaction tubes, and 800 µL chloroform was added to the samples. The samples were mixed vigorously and centrifuged for 1 minute at 10 000 rpm, before transferring the aqueous phase to fresh sterilized 2 mL reaction tubes. 2 µL RNase A [10 mg/ml [ThermoFisher; Cat: EN0531]] were then added, and the samples mixed by inversion and incubated for 15 minutes at 60°C.

Next, 2 volumes of ice-cold 96% ethanol were added to the samples and incubated overnight at -20°C for RNase A inactivation and DNA precipitation. Following overnight incubation, the samples were centrifuged for 20 minutes at 10,000 rpm at 4°C. The supernatants were discarded, and the DNA pellets were washed once in 500 µL 96% ethanol. The samples were then centrifuged for 10 minutes at 10,000 rpm at 4°C, and the DNA pellets washed twice with 500 µL 70% ethanol. After centrifuging for an additional 10 minutes at 10,000 rpm at 4°C, the DNA pellets were dried and resuspended in 100 µL ultra-pure water and incubated overnight at 4°C, before quality assessment. Samples were stored at 4°C.

16*S* rDNA amplicon sequencing and analysis

For bacterial community composition analysis, the V3-V4 fragment of the 16S rDNA gene was amplified from the genomic DNA extracted from the soil and root wash samples. 16S rRNA fragment concentrations in the DNA extracts were quantified using domain-specific quantitative PCR (Fontaine et al., 2023). DNA extracts were normalized based on the 16S rRNA gene concentrations to ensure 16S rRNA gene template molarity for amplification and a two-step barcoding procedure. A first amplification reaction was performed using the 16S rDNA 341F and 805R primer set, containing adapters for the introduction of Illumina adapters (341F + adapter: ACACTCTTTCCCTACACGACGCTCTTCCGATCTNNNNCCTACGGGNGGCWGCAG; 805R + adapter: AGACGTGTGCTCTTCCGATCTGACTACHVGGGTATCTAATCC), using Q5 high-fidelity DNA polymerase [New England Biolabs; Cat: M0491] and the following cycling conditions: 98°C for 1 minute, followed by 20 cycles of 98°C for 10 seconds, 62°C for 30 seconds, 72°C for 30 seconds, and a final extension at 72°C for 2 minutes. The 16S rRNA gene copy concentrations in DNA extracts were determined through quantitative PCR and normalized to equal 16S rRNA fragment copy number to enhance comparability and reduce PCR bias, and purified using the Agencourt AMPure XP [Beckman; Cat: B23319] purification system. Sequencing indexes were introduced in a second amplification reaction with sample-specific barcodes (FWD: AATGATACGGCGACCACCGAGATCTACAC-[index]-ACACTCTTTCCCTACACGACG; REV: CAAGCAGAAGACGGCATACGAGAT-[index]-GTGACTGGAGTTCAGACGTGTGCTCTTCCGATCT) using Q5 high-fidelity DNA polymerase and the following cycling conditions: 98°C for 1 minute, followed by 15 cycles of 98°C for 10 seconds, 66°C for 30 seconds, 72°C for 30 seconds, and a final extension at 72°C for 2 minutes. Amplicon libraries were purified using the Agencourt AMPure XP purification system and quantified using PicoGreen kit [ThermoFisher; Cat: P7589]. The libraries were pooled and diluted to 4 nM before being sequenced for 300 cycles pair-end on the Illumina MiSeq system using the MiSeq® Reagent Kit v2 [Illumina; Cat: MS-102-2002].

Following sequencing, the quality of the 16S amplicon sequences was verified using FastQC (v.0.11.9) (Andrews, 2010) and sequencing adapter sequences, as well as low-quality reads, were removed from the raw read data using Trimmomatic v.0.40 (Bolger et al., 2014) with the following parameters: *ILLUMINACLIP:TruSeq3-PE-2.fa:2:30:10:2:keepBothReads LEADING:3 TRAILING:3 SLIDINGWINDOW:4:15 MINLEN:36*. 16S amplicon sequencing data analysis has been performed using the R package dada2 (v.1.26.0) (Callahan et al., 2016), which includes quality filtering, dereplication, dataset-specific error model determination, ASV interference, chimeral removal, as well as taxonomic assignment, as part of its standard workflow.

Shotgun Sequencing

Shotgun metagenomics sequencing was conducted on the genomic DNA extracted from the soil and root samples. Evaluation of the initial DNA concentrations in the prepared samples was carried out using the Qubit dsDNA HS Assay Kit [ThermoFisher; Cat: Q32851]. The libraries were prepared using 1000 ng of genomic DNA, which was fragmented using the Bioruptor sonication device [Diagenode; Cat: B01020001]. Fragments were cleaned up using the GeneJet PCR Purification Kit [ThermoFisher; Cat. #K0701] and size-selected for a 300 bp using Agencourt

AMPure XP beads. Sequencing libraries were prepared following the standard NEBNext Ultra II DNA Library Prep with Sample Purification Beads [New England Biolabs; Cat: E7103] procedure. The final library concentrations were determined using the Qubit dsDNA HS Assay Kit and the average size of the library was determined using the Agilent 5200 Fragment Analyzer System [Agilent; Cat: M5310AA]. The libraries were pooled and diluted to 4 nM before being sequenced for 600 cycles pair-end on the Illumina MiSeq system using the MiSeq® Reagent Kit v3 [Illumina; Cat: MS-102-3303].

Metagenome assembly, genome binning and quality assessment

The quality of the metagenomic reads was verified using FastQC (v.0.11.9) (Andrews, 2010) and sequencing adapter sequences, as well as low-quality reads, were removed from the sequenced data using Trimmomatic v.0.40 (Bolger et al., 2014) with the following parameters: *ILLUMINACLIP:TruSeq3-PE-2.fa:2:30:10:2:keepBothReads LEADING:3 TRAILING:3 SLIDINGWINDOW:4:15 MINLEN:36*. The quality trimmed reads from both samples were combined into one large dataset, containing the reads from both samples. The combined dataset was assembled using MEGAHIT (v.1.2.9) (Li et al., 2015) (parameters *--k-min 27 --k-max 247 --k-step 10*). Next, the paired-end metagenomic trimmed reads were mapped back to the generated contigs using bowtie2 (v.2.3.5.1) (Langmead and Salzberg, 2012) to generate coverage information, and sorted using SAMtools (v.1.10). To generate MAGs, contigs were binned using Autometa (v2.0) (Miller et al., 2019). Unclustered contigs were succinctly binned using MetaBAT2 (v.17) (Kang et al., 2015). For further improvement, the corresponding reads from each MAGs' contigs were extracted and reassembled using SPAdes (v.3.14.1) (Nurk et al., 2017, Bankevich et al., 2012) using the meta option. Finally, the quality of each MAG was evaluated using CheckM (v.1.1.3) (Parks et al., 2015) and QUAST (v.5.0.2) (Mikheenko et al., 2018), to determine their completeness, contamination and heterogeneity.

MAG analysis and taxonomical classification

For each MAG, ribosomal sequences were extracted based on the KEGG (Kanehisa, 2019, Kanehisa et al., 2023, Kanehisa and Goto, 2000) and PANNZER2 (Toronen et al., 2018) annotations, and quantified by type to identify the most abundant representatives in each MAG. For each binned MAG, taxonomic and lowest common ancestor predictions were generated using Autometa (Miller et al., 2019), based on the non-redundant NCBI database (O'Leary et al., 2016). Furthermore, a library containing genomic sequences of 199 reference organisms (**Supp. Table 1**), obtained from the NCBI database, was generated to calculate the average nucleotide identify (ANI) of each MAG (**Supp. Figure 1**) using FastANI (v.1.33) (Jain et al., 2018).

Gene prediction and annotation

For each generated MAG, genes and proteins were predicted using Prodigal (v.2.6.3) (Hyatt et al., 2010), and the sequences annotated using the KEGG (Kanehisa, 2019, Kanehisa et al., 2023, Kanehisa and Goto, 2000) and PANNZER2 (Toronen et al., 2018) databases. The KEGG annotation was further used to investigate metabolic pathways, whereas PANNZER2 was used to annotate gene ontology and functional domains. Additionally, biosynthetic gene clusters (BGCs) for each MAG were predicted using antiSMASH (v.6.1.1) (Blin et al., 2017, Blin et al., 2021), DeepBGC (v.0.1.31) (Hannigan et al., 2019), and GECCO (v.0.9.10) (Carroll et al., 2021), annotated using the KEGG and PANNZER2 databases and evaluated for completeness.

Results

16S rRNA gene amplicon sequencing/taxonomic identification

To gain insights into the rhizosphere of *D. traunsteineri*, a potted orchid (Individuum 1 from Kitzbühel, sampled on 25.05.2018), maintained in the botanical garden of the University of Vienna (Universität Wien), has been transferred to our laboratory environment (**Figure 1**). To taxonomically characterize the predominant bacterial and archaeal community inhabiting the

Samples	SRA accession no.	No. of raw sequence reads	No. of reads that passed quality control	ASVs after chimera removal
Soil-R1	SRR29434315	36,584	25,981	14,743
Soil-R2	SRR29434314	95,543	68,635	45,317
Wash-R1	SRR29434311	29,562	20,461	11,303
Wash-R2	SRR29434310	79,246	55,090	35,285

Table 1. **Sequencing statistics (16S profiling)**

rhizosphere of *D. traunsteineri*, we conducted PCR amplification and sequencing of the V3-V4 variable region of the 16S rRNA gene from the soil and root wash samples in technical duplicates. Following paired-end sequencing, a total of 240,935 reads were generated, each with a read length of 300 bp (**Table 1**). After rigorous processing steps, including trimming, filtering of low-quality reads, merging of forward and reverse reads, and removal of chimeric sequences, 106,648 sequences were clustered into 2350 amplicon sequence variants (ASV) with taxonomic assignments spanning 31 phyla and over 84 classes (**Figure 2**). In total, we identified 2349 bacterial and 1 archaeal ASVs. The prevalence of the five major bacterial phyla ranged from 38.43% to 41.33% for *Proteobacteria*, 11.09% to 13.55% for *Actinobacteriota*, 9.40% to 11.90% for *Myxococcota*, 9.72% to 11.37% for *Bacteriodota*, and from 8.72% to 9.26% for *Acidobacteriota* (**Table 2**). Comparisons of ASV abundances across the four samples using permutational multivariate analysis of variance (PERMANOVA) with the R package vegan revealed no significant differences (similarity p-value < 0.09), indicating a consistent sampling and sequencing approach. Encouraged by the stability of the bacterial composition in the soil surrounding the roots of *D. traunsteineri* and the bacteria adhering to the roots themselves (root wash samples), we opted to analyze and process the shotgun metagenomic sequences as a unified sample.

Metagenome assembled genomes (MAGs) generation

To investigate the functional properties of the *D. traunsteineri* rhizobiome, we proceeded to shotgun metagenomic sequencing using short-read technology. We generated 33,043,213 paired-end metagenomic reads from the soil and root wash samples, totalling over 9 Gbp, with each sample ranging from over 1.8 to over 4.9 Gbp (**Table 3**). After in silico pruning of adapter sequences and low-quality reads, 31,541,795 reads were retained, representing over 95% of the initial dataset. *De novo* metagenomic assembly on the combined datasets using MEGAHIT (Li et al., 2015, Li et al., 2016) resulted in a total of 859,083 contigs, totalling over 7.5 Gbp (**Table 4**).

The obtained contigs were binned using Autometa (Miller et al., 2019), resulting in 37 predicted MAGs, as well as a group of unclustered contigs, which was further binned using MetaBAT2 (Kang et al., 2015), thus resulting in 10 additional MAGs (**Figure 3**). Overall, from the 248,298 contigs that were binned as Bacteria, 44,138 contigs could be grouped into 47 MAGs, from which 36 MAGs could successfully be reassembled using SPAdes (Bankevich et al., 2012, Nurk et al., 2017). These MAGs represent 359 Mbp of assembled reads, while 14.86% of all reads mapped back to the sequences contained in the combined bacterial MAGs (**Table 5**). MAG size ranges from 361 kbp to 45.81 Gbp, with N50 ranging from 0.9 kbp to 221 kbp (Table 6). The average predicted completeness of the MAGs stands at 33.64%, with a mean coverage spanning from 2.11 to 25.00x. Following a quality assessment, 13 initially generated MAGs were excluded due to either a contamination score exceeding 50% or completeness falling below 10%. The taxonomic distribution of the top-represented groups revealed that 28 MAGs were classified as *Proteobacteria*, eight as *Acidobacteria*, and eight as *Bacteroidetes* (Table 7). Notably, our assembly efforts yielded one high-quality draft genome (MAG_Dt_26), satisfying the stringent Minimum Information about a Metagenome-Assembled Genome (MIMAG) criteria, boasting less than 5% contamination and over 90% completeness, as well as three medium-quality draft genomes (MAG_Dt_01, MAG_Dt_03, MAG_Dt_23), with estimated completeness above 50% with less than 10% estimated contamination (Bowers et al., 2017).

Gene prediction and secondary metabolite potential

Gene prediction was performed using PRODIGAL (Hyatt et al., 2010) for each MAG, and the resulting predicted proteins were annotated using the KEGG (Kanehisa and Goto, 2000, Kanehisa et al., 2023) and PANNZER2 (Toronen et al., 2018) databases for metabolic pathway enrichment (Table 8). Furthermore, each MAG was mined for SM BSGs using the nf-core funcscan pipeline (Ewels et al., 2020). A total of 1,741 BGCs were predicted from the *D. traunsteineri* rhizosphere bacterial MAGs (**Figure 4**; Table 9).

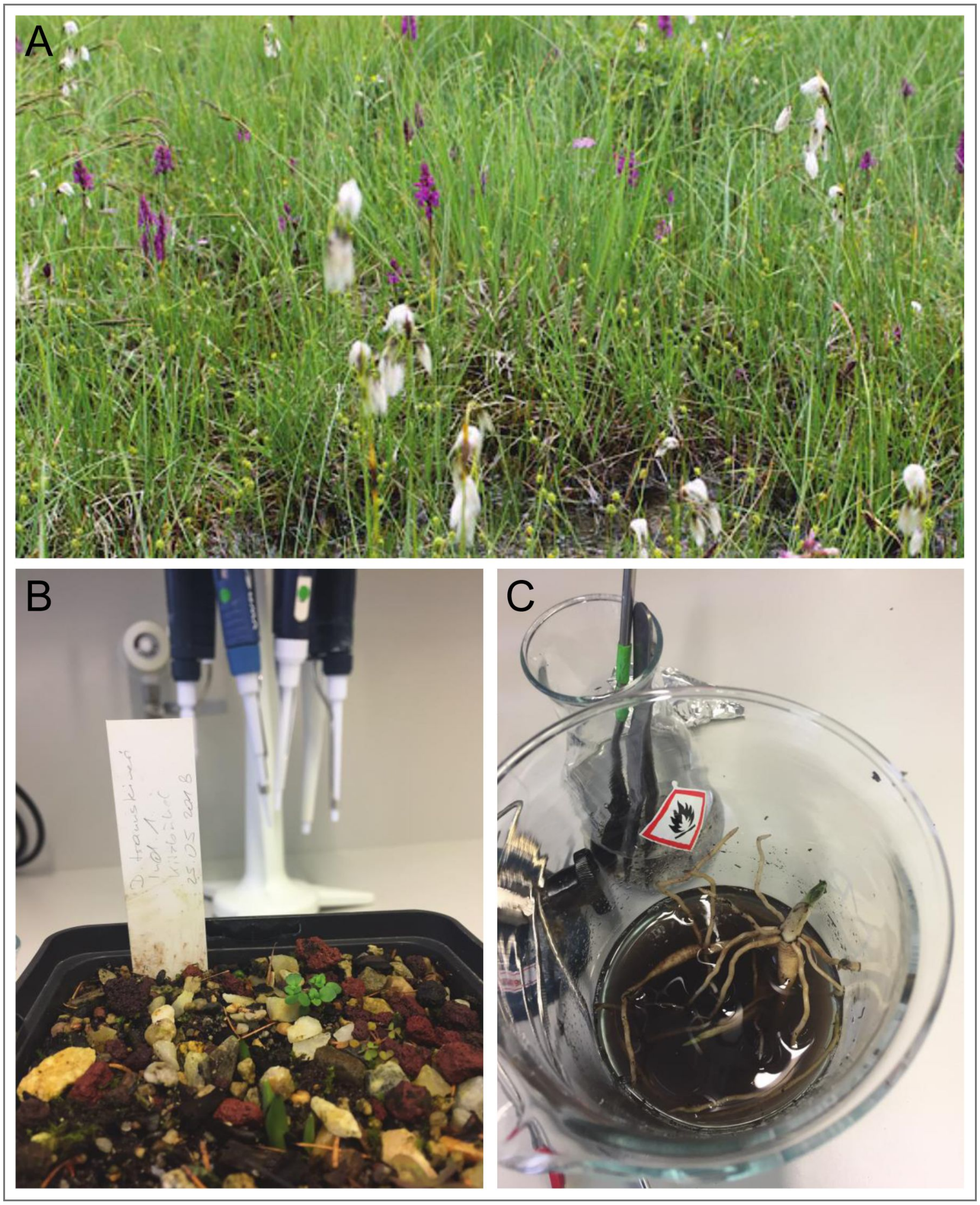


Figure 1. ***Dactylorhiza traunsteineri* sampling.**

D. traunsteineri individuals from the Kitzbühel region (Tyrol, Austria) (**A**). *D. traunsteineri* individuum 1 from Kitzbühel in the laboratory environment (**B**). Washing of the roots with laboratory-grade water (**C**).

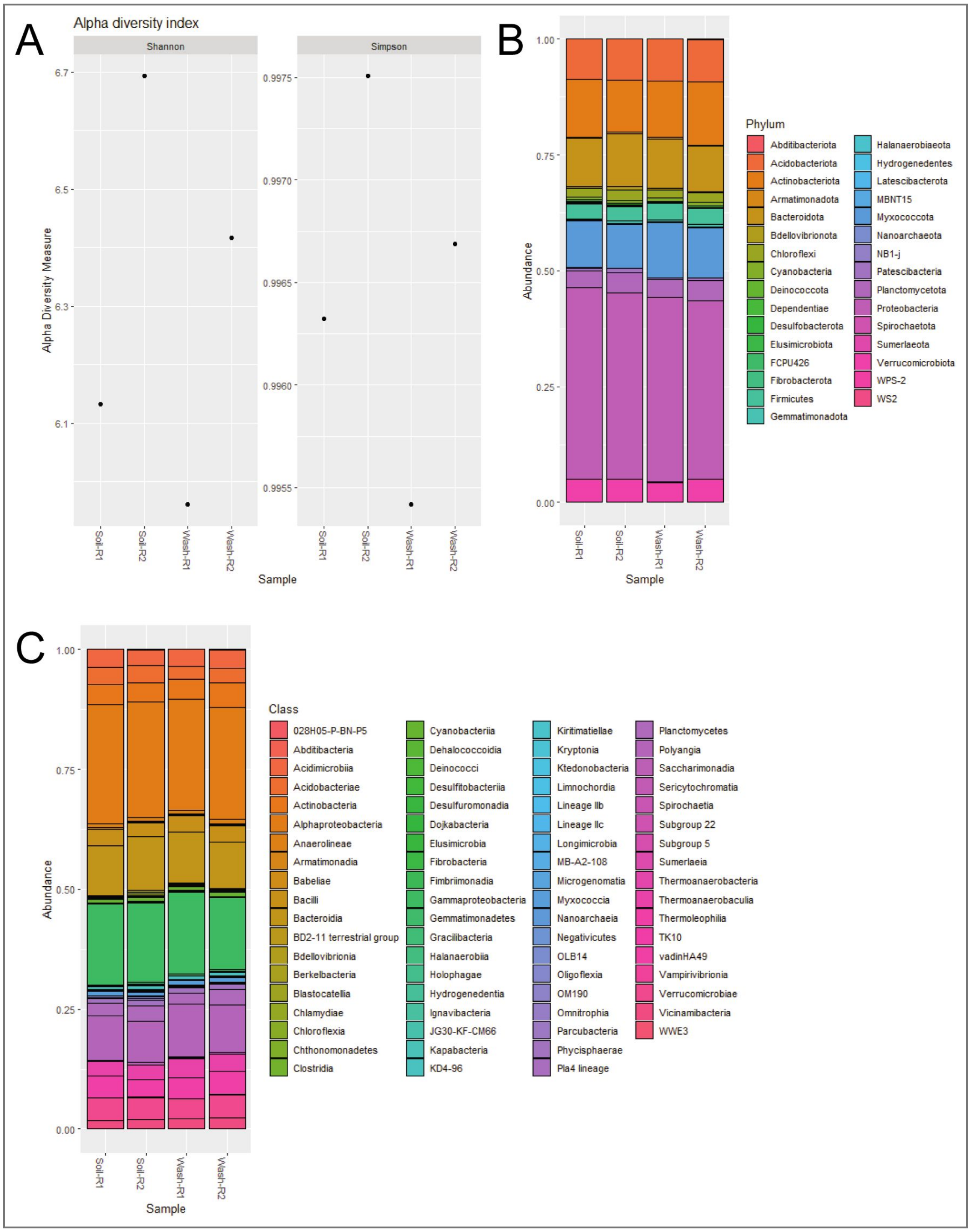


Figure 2. **Bacterial Community Composition Analysis based on 16S rDNA sequencing.**

Alpha-diversity (Shannon and Simpson indexes) (**A**). Relative abundance of identified ASVs at Phylum (**B**) and Class (**C**) levels.

Table 2. ASVs abundance (by Phylum) (%)

	Soil-R1	Soil-R2	Wash-R1	Wash-R2
Proteobacteria	41.33	40.11	39.98	38.43
Actinobacteriota	12.41	11.09	12.20	13.55
Myxococcota	10.15	9.40	11.90	10.84
Bacteroidota	10.40	11.37	10.56	9.72
Acidobacteriota	8.72	8.85	9.02	9.26
Verrucomicrobiota	4.92	5.04	4.29	4.97
Planctomycetota	3.59	4.36	3.64	4.33
Firmicutes	3.34	2.88	3.54	3.40
Chloroflexi	1.82	2.24	1.87	2.03
Cyanobacteria	0.65	0.66	0.70	0.84
Patescibacteria	0.54	1.02	0.40	0.66
Gemmatimonadota	0.24	0.57	0.41	0.48
Bdellovibrionota	0.41	0.70	0.27	0.28
Armatimonadota	0.22	0.48	0.41	0.29
Dependentiae	0.35	0.34	0.17	0.31
Desulfobacterota	0.31	0.15	0.09	0.14
Latescibacterota	0.11	0.13	0.12	0.09
FCPU4260	0.07	0.12	0.12	0.07
Elusimicrobiota	0.12	0.09	0.08	0.06
MBNT15	0	0.10	0.13	0.07
Fibrobacterota	0.01	0.13	0.04	0.01
Spirochaetota	0.03	0.01	0.12	0.04
NB1-j	0.05	0.02	0.04	0.02
WPS-2	0.09	0.01	0	0
Sumerlaeota	0.03	0.03	0	0.02
Nanoarchaeota	0.05	0	0	0
Abditibacteriota	0	0.01	0.02	0.02
WS2	0	0.01	0	0.04
Hydrogenedentes	0	0.02	0	0.01
Deinococcota	0	0.01	0	0
Halanaerobiaeota	0	0.01	0	0

Table 3. Sequencing statistics (Shotgun sequencing)

Samples	SRA accession no.	No. of raw sequence reads	No. of reads that passed quality control
Soil	SRR29434313	6,252,114	6,050,295
Wash	SRR29434312	10,152,392	9,850,373

Table 4. **Metagenomic assembly statistics**

No. of contigs	859,083				
Largest contig	368,968 bp				
Total length	757,260,241 bp				
N_{50}	829				
L_{50}	252795				
Bins (MAGs)	47	0	0	7	41

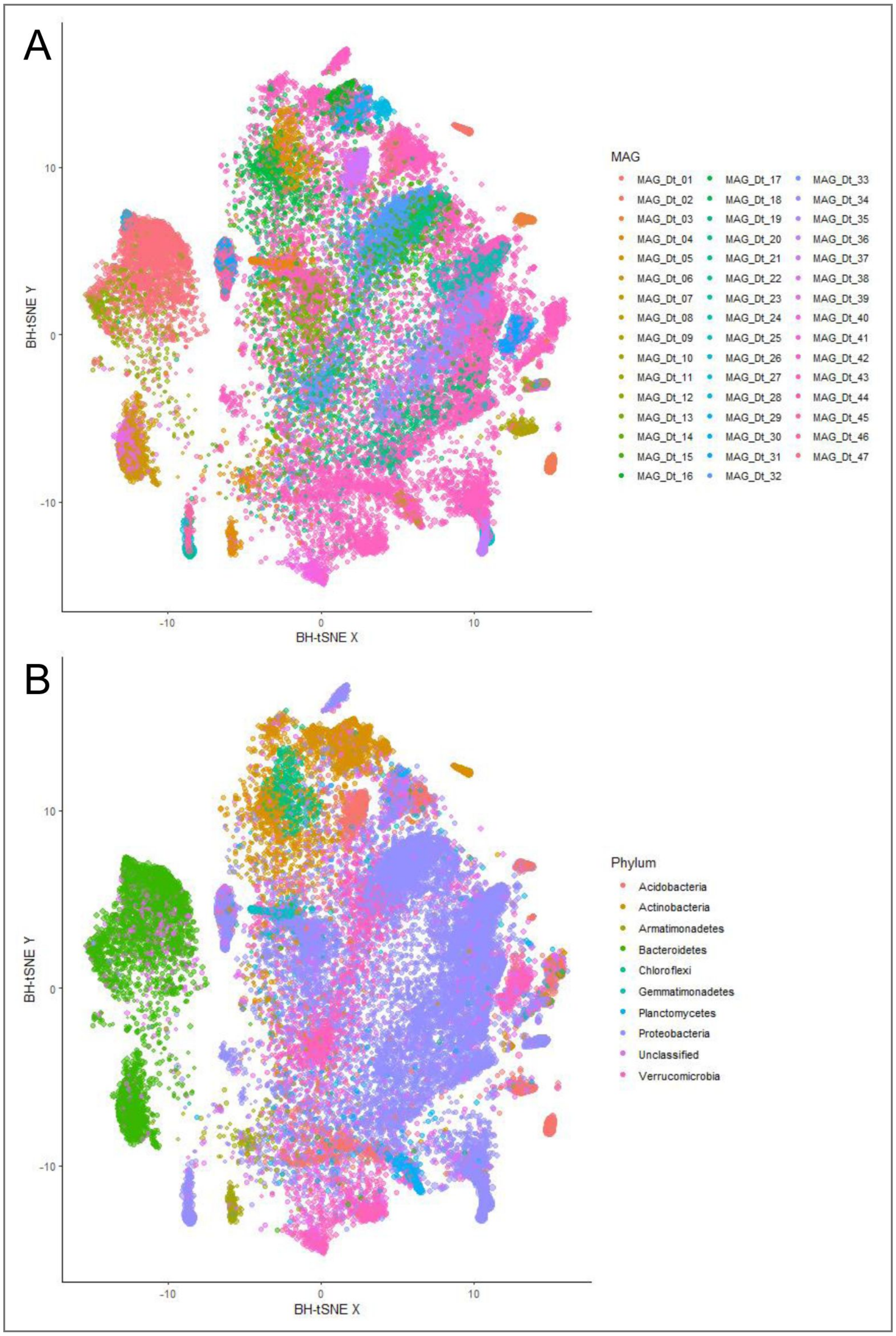


Figure 3. Barnes-Hut t-SNE representation of MAG contigs.

BH-t-SNE representation of MAG contigs coloured by MAG (**A**) and taxonomic assignment (**B**).

	Bacteria	Archaea	Eukaryota	Virus	Plasmid
No. of reads	4,909,848	1,308	23,299	6,397	4,894
Percentage (%)	14.86%	0.004%	0.07%	0.01%	0.01%
No. of contigs	248,298	87	2,289	148	41
Largest contig (bp)	368,968	5,444	23,737	28,310	45,246
Total length (bp)	366,412,546	105,809	2,577,161	376,352	88,614
N_{50}	1,372	1,155	1,046	3,945	45,246
L_{50}	64,708	32	860	23	1

Table 5. **Metagenomic binning statistics**

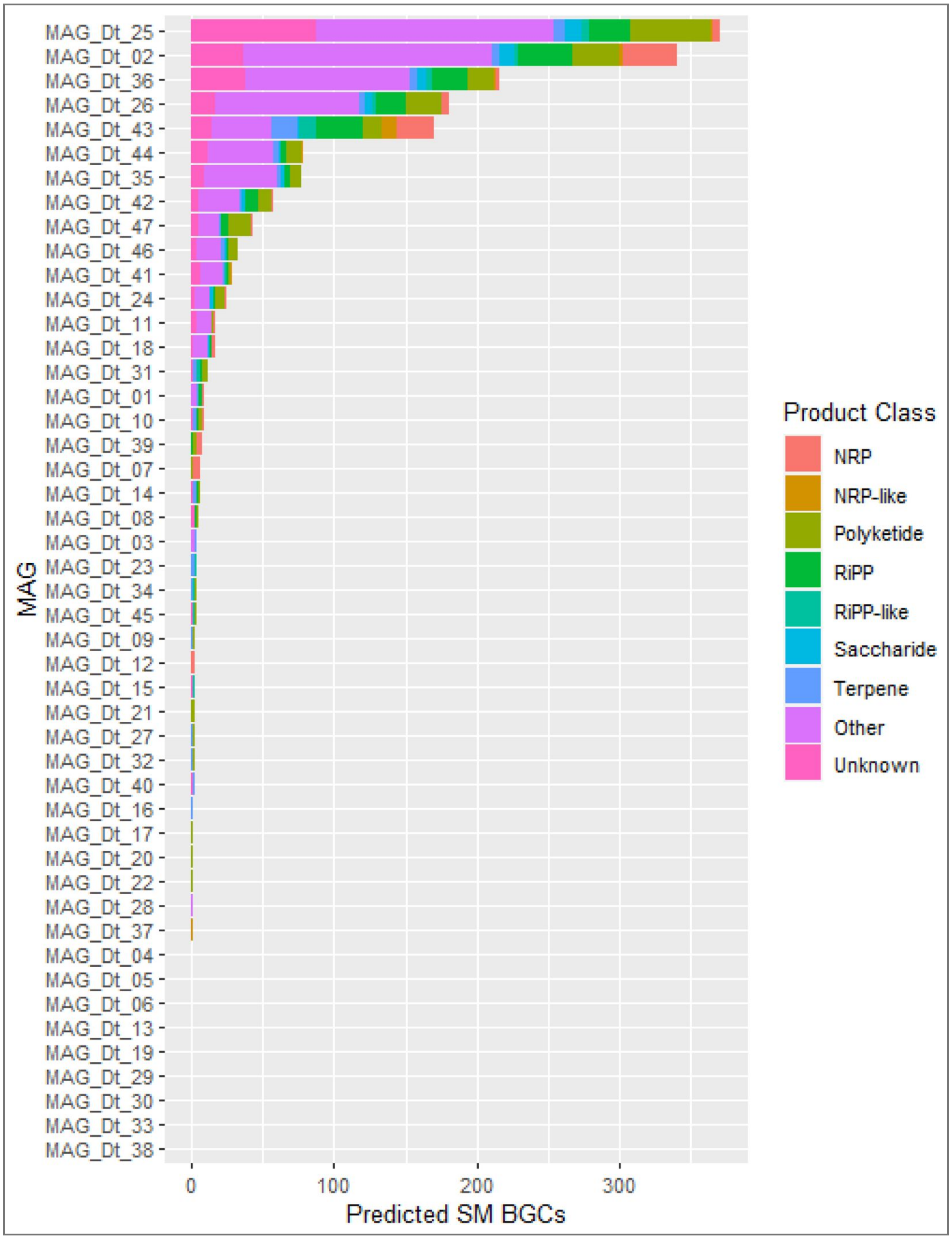


Figure 4. **Overview of number of predicted SM BGCs per MAG and their product classes.**

In MAG_Dt_26 (*Proteobacteria*; completeness of 92.03%; contamination of 2.53%), a total of 181 SM BGCs were predicted. These included six nonribosomal peptide (NRP) clusters, 24 polyketide clusters, 22 ribosomally synthesized and post-translationally modified peptide (RiPP) clusters, two RiPP-like clusters, five saccharide clusters, as well as four terpene clusters (Supp. Table 2). Additionally, 101 gene clusters were categorized as "Other", potentially representing multiple product types and rarer SM classes, while 17 clusters were classified as "Unknown" (Supp. Table 2). From MAG_Dt_01 (*Actinobacteria*; completeness of 80.24%; contamination of 4.58%), a total of nine SM BGCs were predicted, representing one NRP cluster, one polyketide cluster, two RiPP clusters, two terpene clusters, as well as three "Other" product class clusters (Supp. Table 2). From MAG_Dt_03 (*Acidobacteria*; completeness of 65.94%; contamination of 3.99%), four SM BGCs were predicted, representing one NRP cluster, one terpene cluster, and two "Other" product class clusters (Supp. Table 2). From MAG_Dt_23 (*Proteobacteria*; completeness of 61.01%; contamination of 2.68%), four SM BGCs were predicted, representing two RiPP-like and two terpene clusters (Supp. Table 2).

Discussion

Through comprehensive 16S rDNA profiling, a total of 2,349 bacterial and one archaeal ASVs were identified within the rhizosphere of *D. traunsteineri*. The predominant bacterial phyla detected included *Proteobacteria*, *Actinobacteria*, *Myxococcota*, *Bacteroidota*, and *Acidobacteria*, which collectively accounted for 95% of the sequences identified. Importantly, there were no significant differences in the bacterial community composition between the soil and the root-adhering bacteria associated with *D. traunsteineri*. This consistency underscores the robustness and reproducibility of our sample preparation and sequencing methodologies. Moreover, in our review of existing literature on orchid RAB ecology, we validated that our approach successfully recovered representatives from all phyla previously reported in orchid rhizospheres (Kaur and Sharma, 2021). Notably, the scientific documentation of orchid RAB through culture-independent methods is relatively scarce. Previous studies have predominantly utilized conventional Sanger sequencing methods, while others employing high-throughput sequencing technologies have primarily focused on 16S rDNA profiling of bacterial communities directly extracted from root samples. Nevertheless, the field of orchid RAB research is undergoing a transformative shift with the advent of innovative methodologies. In this context, we introduce a pioneering approach that leverages shotgun metagenomics, followed by the reconstruction of MAGs. This cutting-edge technique promises to offer deeper insights into the complex ecosystem of the orchid rhizobiome, providing unparalleled understanding of its microbial diversity and functional potential. Utilizing shotgun sequencing and *de novo* metagenomic assembly, we successfully obtained 47 bacterial MAGs from the rhizobiome of *D. traunsteineri.* Our detailed analysis focused on four high-quality MAGs: MAG_Dt_01, MAG_Dt_03, MAG_Dt_23, and MAG_Dt_26. Specifically, MAG_Dt_01 was taxonomically assigned to *Actinobacteria bacterium* (taxid 1883427), MAG_Dt_03 to *Acidobacteria bacterium* (taxid 1978231), and MAG_Dt_23 and MAG_Dt_26 to *Dongia* sp. URHE0060 (taxid 1380391) and *Steroidobacter* (taxid 469322), respectively, both belonging to the phylum *Proteobacteria*.

The phyla *Proteobacteria* and *Actinobacteria*, generally enriched in the rhizosphere, alongside *Acidobacteria*, typically enriched in bulk soil, play pivotal roles in soil carbon and nitrogen cycles as well as the decomposition of organic matter (Eichorst et al., 2018, Song et al., 2021, Ling et al., 2022). The *Actinobacteria* phylum represents one of the largest and most diverse bacterial group, inhabiting soil, water and plant tissue (Diab et al., 2024, Boubekri et al., 2022), and are prolific producers of SMs including well-known antibiotics such as streptomycin, tetracycline, erythromycin, and vancomycin (Berdy, 2005). Beyond antibiotics, they produce a wide array of SMs with potential applications as biopesticides (Barka et al., 2016). For example, polyketides, such as macrolides, exhibit antifungal activity, while non-ribosomal peptides (NRPs), like cyclopeptides, demonstrate insecticidal properties (Gaynor and Mankin, 2003, Quiroz-Carreno et al., 2022). The *Acidobacteria* phylum, although highly prevalent within soil ecosystems, is characterized by its challenging cultivation (Kalam et al., 2020). This phylum fulfills crucial ecological functions through its significant involvement in carbon, nitrogen, and sulphur biogeochemical processes (Ward et al., 2009, Eichorst et al., 2018, Hausmann et al., 2018). Furthermore, *Acidobacteria* possess genes conducive to survival and competitive colonization within the rhizosphere, thereby facilitating the formation of symbiotic relationships with plants (Kalam et al., 2020, Liu et al., 2024). Members of the *Proteobacteria* phylum, such as the *Dongia* and *Steroidobacter* genera, have been frequently isolated from rhizosphere soil and display the potential to degrade complex organic compounds, including various polysaccharides (Ikenaga et al., 2021, Huang et al., 2019, Liu et al., 2024, Baik et al., 2013, Kim et al., 2016, Liu et al., 2010, Palla et al., 2022). The high-quality MAGs obtained in this study present a promising opportunity to elucidate in greater detail the roles of these phyla in plant microbiome interactions, particularly regarding their secondary metabolite potential and their impact on plant health and development.

Given that our sampling and sequencing efforts were confined to the rhizobiome of a solitary specimen of *D. traunsteineri*, caution must be exercised in generalizing the presence and prevalence of the identified MAGs across all populations of *D. traunsteineri*. However, it is important to note that the sampled individual originated from an extant population and has been maintained under conditions that closely simulate its natural habitat at the Botanical Garden of the University of Vienna. These conditions include soil composition, humidity, and temperature, which are critical for mimicking the plant's natural environment. The controlled yet naturalistic setting in which the specimen was studied provides a valuable context for our findings. Despite the inherent limitations of a single-sample study, our research offers a foundational framework for future interaction studies between RABs and *D. traunsteineri*. This study lays the groundwork for more extensive investigations into the microbiome of this orchid species across different populations and environmental conditions. Furthermore, our findings open new avenues for exploring the complex processes involved in the adaptation, ecology, and evolution of this allopolyploid orchid. Understanding the interactions between *D. traunsteineri* and its associated microbiome can shed light on the mutualistic relationships that facilitate plant health, nutrient acquisition, and stress resilience. Such insights are crucial for conservation strategies, especially for orchids that are often sensitive to environmental changes and habitat disturbances. Additionally, the use of advanced metagenomic techniques in our study underscores the potential for discovering novel microbial taxa and functional genes that contribute to the unique ecology of orchid rhizospheres. This approach not only enhances our understanding of plant-microbe interactions but also provides a deeper comprehension of the evolutionary dynamics within the orchid family. Our study, therefore, serves as a stepping stone for future research aimed at unraveling the intricate web of interactions that sustain and influence the biodiversity and ecological success of *D. traunsteineri*.

Conclusion

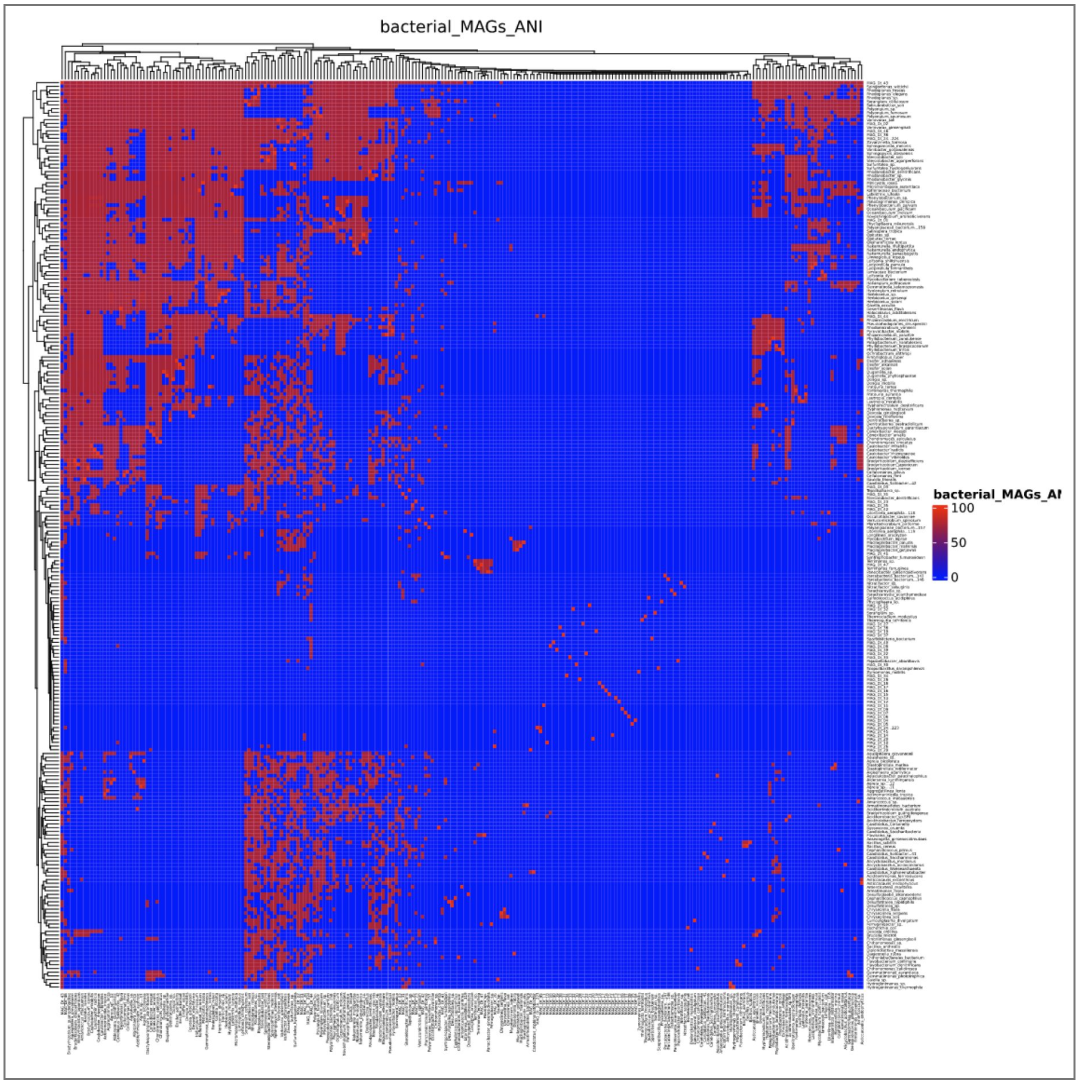
This study provided a comprehensive overview of the microbiome inhabiting the rhizosphere of the endangered orchid *Dactylorhiza traunsteineri*. By conducting an in-depth bacterial community composition analysis through 16S rDNA-targeted sequencing, we identified *Proteobacteria*, *Actinobacteria*, *Myxococcota*, *Bacteroidota*, and *Acidobacteria* as the most abundant phyla within the *D. traunsteineri* rhizosphere. This foundational data highlights the complexity and diversity of the microbial communities associated with this orchid species. Our investigation extended beyond community profiling to include deep shotgun metagenomics sequencing and subsequent *de novo* metagenome assembly. This approach enabled us to extract high-quality metagenome-assembled genomes (MAGs), providing a rich resource for further functional analysis. Each MAG was meticulously analyzed for metabolic pathway enrichment and secondary metabolite (SM) biosynthetic gene clusters (BGCs), revealing a previously unexplored niche of the rhizobiome. These analyses uncovered potential metabolic capabilities and biosynthetic potentials that are integral to the rhizosphere's ecological dynamics.

The insights gained from this study not only enhance our understanding of the microbial diversity within the *D. traunsteineri* rhizosphere but also lay the groundwork for future research aimed at unravelling the intricate interconnections between the rhizobiome and its host. The high-quality MAGs we have made available serve as a valuable resource for further exploration of microbial functions and their contributions to the health, adaptation, and survival of *D. traunsteineri*. Moreover, our findings underscore the importance of employing advanced metagenomic techniques to study plant-associated microbiomes, particularly for endangered species. By elucidating the functional roles of key microbial players, we can develop more effective conservation strategies and foster a deeper understanding of the symbiotic relationships that underpin plant resilience and ecological success. In conclusion, our study represents a significant step forward in orchid microbiome research, providing novel insights and a robust platform for future investigations into the dynamic interactions within the rhizosphere of *D. traunsteineri*. These efforts are crucial for advancing our knowledge of plant-microbe interactions and supporting the conservation of this and other endangered orchid species.

Software

Software	Version	Reference
antiSMASH	v.6.1.1	(Blin et al., 2017, Blin et al., 2021)
Autometa	v.2.0	(Miller et al., 2019)
bowtie2	v.2.3.5.1	(Langmead and Salzberg, 2012)
CheckM	v.1.1.3	(Parks et al., 2015)
dada2 (R package)	v.1.16.0	(Callahan et al., 2016)
DeepBGC	v. 0.1.31	(Hannigan et al., 2019)
FastANI	v.1.33	(Jain et al., 2018)
FastQC	v.0.11.9	(Andrews, 2010)
funcscan	v.1.1.5	(Ewels et al., 2020)
GECCO	v.0.9.10	(Carroll et al., 2021)
ggtree (R package)	v.3.0.4	(Yu, 2020, Yu et al., 2018, Yu et al., 2016)
MEGAHIT	v.1.2.9	(Li et al., 2016, Li et al., 2015)
MetaBAT2	v.17	(Kang et al., 2015)
OrthoFinder	v.2.5.2	(Emms and Kelly, 2019)
Prodigal	v.2.6.3	(Hyatt et al., 2010)
QUAST	v.5.0.2	(Mikheenko et al., 2018)
SAMTools	v.1.10	(Li et al., 2009)
SPAdes	v.3.14.1	(Bankevich et al., 2012, Nurk et al., 2017)
Trimmomatic	v.0.40	(Bolger et al., 2014)

Supplementary Information



Supplementary Figure 1. **ANI of MAGs and reference organisms.**

Genome	GenBank Assembly	Reference
Acidiferrimicrobium australe strain USS-CCA1	GCA_009697115.1	(Gonzalez et al., 2020)
Acidiferrobacter sp. SPIII/3	GCA_003184265.1	N/A
Acidihalobacter ferrooxydans strain V8	GCA_001975725.1	(Khaleque et al., 2019)
Aciditerrimonas ferrireducens JCM 15389	GCA_001311945.1	(Yamazaki et al., 2010)
Actinomarinicola tropica sp. nov.	GCA_009650215.1	(He et al., 2020b)
Aggregatilinea lenta sp. nov.	GCA_003569045.1	(Nakahara et al., 2019)
Agreia bicolorata strain AC-1804	GCA_000938265.1	(Evtushenko et al., 2001)
Agreia sp. COWG	GCA_904528075.1	N/A
Agreia sp. Leaf244	GCA_001422695.1	(Bai et al., 2015)
Aldersonia kunmingensis strain DSM 45001	GCA_001646865.1	(Sangal et al., 2016)
Algisphaera agarilytica strain DSM 103725	GCA_014207595.1	(Kyrpides et al., 2014)
Alicyclobacillus acidocaldarius subsp. acidocaldarius DSM 446	GCA_000024285.1	(Mavromatis et al., 2010)
Alicyclobacillus montanus strain USBA-GBX-503	GCA_900142255.1	(Lopez et al., 2018)
Amaricoccus macauensis strain DSM 101730	GCA_014201905.1	(Kyrpides et al., 2014)
Amaricoccus solimangrovi strain HB172011	GCA_006385685.1	(Mo et al., 2020)
Anseongella ginsenosidimutans strain Gsoil 524	GCA_008033235.1	(Siddiqi et al., 2016)
Aquidulcibacter paucihalophilus strain TH1-2	GCA_002105465.1	(Cai et al., 2017)
Aquisphaera insulae strain JC669	GCA_011064615.1	(Kumar et al., 2021)
Aquisphaera giovannonii strain OJF2	GCA_008087625.1	(Bondoso et al., 2011)
Ardenticatena maritima strain 110S	GCA_001306175.1	(Kawaichi et al., 2013)
Armatimonadetes bacterium Uphvl-Ar1	CP021423.1	(Woodhouse et al., 2017)
Armatimonas rosea strain DSM 23562	GCA_014202505.1	(Tamaki et al., 2011)
Asticcacaulis endophyticus strain KCTC 32296	GCA_014652575.1	(Zhu et al., 2014)
Asticcacaulis excentricus CB 48	GCA_000175215.2	(Liu et al., 2005)
Bacillus anthracis strain Ames	GCA_000007845.1	(Read et al., 2003)
Bacillus cereus ATCC 14579	GCA_000007825.1	(Ivanova et al., 2003)
Bacillus subtilis subsp. subtilis strain 168	GCA_000009045.1	(Kunst et al., 1997)
Bauldia litoralis ATCC 35022	GCA_900104485.1	(Yee et al., 2010)
Blastopirellula marina DSM 3645	GCA_000153105.1	(Schlesner et al., 2004)
Blastopirellula retiformator	GCA_007859755.1	(Kallscheuer et al., 2020)
Bradyrhizobium diazoefficiens USDA 110	GCA_000011365.1	(Kaneko et al., 2002)
Bradyrhizobium guangdongense strain CCBAU 51649	GCA_004114975.1	N/A
Bradyrhizobium icense strain LMTR 13	GCA_001693385.1	(Ormeno-Orrillo et al., 2018)
Bradyrhizobium japonicum USDA 6	GCA_000284375.1	(Kaneko et al., 2011)
Brucella microti CCM 4915	GCA_000022745.1	(Audic et al., 2009)
Byssovorax cruenta JCM 12614	GCA_001312805.1	N/A
Candidatus Carsonella ruddii isolate JRPAMB4	GCA_013463375.1	N/A
Candidatus Protochlamydia phocaeensis strain C2	GCA_001545115.1	N/A
Candidatus Saccharibacteria bacterium RAAC3_TM7_1	GCA_000503915.1	(Kantor et al., 2013)
Candidatus Saccharimonas aalborgensis isolate TM71	GCA_000392435.1	(Albertsen et al., 2013)
Candidatus Solibacter sp.	GCA_011375985.1	N/A
Candidatus Solibacter usitatus Ellin6076	GCA_000014905.1	N/A
Candidatus Woesearchaeota archaeon	GCA_016432305.1	(He et al., 2020a)

Supplementary Table 1. Genomes used for comparative analysis

Candidatus Xiphinematobacter sp. Idaho Grape	GCA_001318295.1	(Brown et al., 2015)
Caulobacter mirabilis strain FWC 38	GCA_002749615.1	N/A
Caulobacter radicis strain 695	GCA_003094615.1	N/A
Caulobacter rhizosphaerae strain KCTC 52515T	GCA_010977555.1	N/A
Caulobacter crescentus CB15	GCA_000006905.1	(Nierman et al., 2001)
Cellulomonas fimi ATCC 484	GCA_000212695.1	N/A
Cellulomonas gilvus ATCC 13127	GCA_000218545.1	(Christopherson et al., 2013)
Cephaloticoccus capnophilus strain CV41	GCA_001580045.1	N/A
Cephaloticoccus primus strain CAG34	GCA_001580015.1	N/A
Chitinophaga rupis strain DSM 21039	GCA_900109685.1	N/A
Chondromyces apiculatus DSM 436	GCA_000601485.1	N/A
Chondromyces crocatus strain Cm c5	GCA_001189295.1	(Zaburannyi et al., 2016)
Chryseolinea flava strain SDU1-6	GCA_003286915.1	(Wang et al., 2018)
Chryseolinea serpens strain DSM 24574	GCA_900129725.1	N/A
Chryseolinea soli strain KIS68-18	GCA_003589925.1	(Lee et al., 2019b)
Chthoniobacterales bacterium	GCA_003244145.1	(Ji et al., 2017)
Chthonomonas calidirosea T49	GCA_000427095.1	(Lee et al., 2014)
Chthonomonas sp. UBA2362	GCA_002344135.1	(Parks et al., 2017)
Conexibacter arvalis strain DSM 23288	GCA_014199525.1	N/A
Conexibacter woesei DSM 14684	GCA_000025265.1	(Pukall et al., 2010)
Cuniculiplasma divulgatum strain PM4	GCA_900090055.1	(Golyshina et al., 2016)
Dactylosporangium aurantiacum strain NRRL B-8018	GCA_000716715.1	(Doroghazi et al., 2014)
Daejeonella rubra strain DSM 24536	GCA_900103545.1	N/A
Denitratisoma oestradiolicum strain DSM16959	GCA_902813185.1	N/A
Denitratisoma sp. DHT3	GCA_007833355.1	(Wang et al., 2020)
Desertimonas flava strain SYSU D60003	GCA_003426815.1	(Asem et al., 2018)
Desulfatitalea sp. UBA4791	GCA_002402905.1	(Parks et al., 2017)
Desulfatitalea tepidiphila strain S28bF	GCA_001293685.1	(Higashioka et al., 2013)
Desulfoglaeba alkanexedens ALDC	GCA_005377625.1	(Kropp et al., 2000)
Devosia crocina strain IPL20	GCA_900116545.1	N/A
Devosia ginsengisoli strain Gsoil 520	GCA_007859655.1	(Quan et al., 2020)
Devosia riboflavina strain IFO13584	GCA_000743575.1	(Hassan et al., 2014)
Diplorickettsiella massiliensis 20B	GCA_000257395.1	(Mathew et al., 2012)
Dongia mobilis strain CGMCC 1.7660	GCA_004363235.1	(Whitman et al., 2015)
Dongia sp. URHE0060	GCA_000620685.1	N/A
Duganella phyllosphaerae strain T54	GCA_001758785.1	(Haack et al., 2016)
Duganella dendranthematis strain AF9R3	GCA_012849375.1	N/A
Ensifer adhaerens strain Casida A	GCA_000697965.2	(Williams et al., 2017)
Ensifer alkalisoli strain YIC4027	GCA_008932245.1	(Dang et al., 2019)
Ensifer sojae CCBAU 05684	GCA_002288525.1	N/A
Escherichia coli str. K-12 substr. MG1655	GCA_000005845.2	(Blattner et al., 1997)
Fimbriiglobus ruber strain SP5	GCA_002197845.1	(Ravin et al., 2018)
Fimbriimonas ginsengisoli Gsoil 348	GCA_000724625.1	(Hu et al., 2014)
Flavitalea sp. OAE831	GCA_014873345.1	N/A
Flavobacterium commune strain PK15	GCA_001857965.1	N/A

Supplementary Table 1. (continued)

Flavobacterium denitrificans DSM 15936	GCA_000425445.1	N/A
Fontimonas thermophila strain DSM 23609	GCA_900113085.1	N/A
Frateuria aurantia DSM 6220	GCA_000242255.3	N/A
Frateuria terrea strain DSM 26515	GCA_900109025.1	N/A
Gaiella occulta strain F2-233	GCA_003351045.1	(Severino et al., 2019)
Gaiella sp. SCGC AG-212-M14	GCA_001644475.1	(Cabello-Yeves et al., 2018)
Gemmatimonas aurantiaca T-27	GCA_000010305.1	(Zhang et al., 2003)
Gemmatimonas phototrophica strain AP64	GCA_000695095.2	(Zeng et al., 2016)
Gemmatirosa kalamazoonesis strain KBS708	GCA_000522985.1	(Debruyn et al., 2014)
Haliangium ochraceum DSM 14365	GCA_000024805.1	(Ivanova et al., 2010)
Halocalculus aciditolerans strain JCM 19596	GCA_014647475.1	(Wu and Ma, 2019)
Hanamia caeni strain BO-59	GCA_003721595.1	(Choi et al., 2022)
Herbiconiux ginsengi CGMCC 4.3491	GCA_900107435.1	N/A
Herbiconiux solani NBRC 106740	GCA_001571005.1	(Behrendt et al., 2011)
Herbiconiux sp. SALV-R1	GCA_013113715.1	N/A
Hyalangium minutum DSM 14724	GCA_000737315.1	N/A
Hydrogenimonas sp.	GCA_003945285.1	(Mino et al., 2018)
Hydrogenimonas thermophila EP1-55-1	GCA_900115615.1	N/A
Hyphomicrobium denitrificans ATCC 51888	GCA_000143145.1	N/A
Hyphomonas neptunium ATCC 15444	GCA_000013025.1	(Badger et al., 2006)
Kofleriaceae bacterium H4 MYX2	GCA_015075455.1	(Liu et al., 2020)
Labilithrix luteola strain DSM 27648	GCA_001263205.1	N/A
Lacipirellula limnantheis strain I41	GCA_007746075.1	(Kallscheuer et al., 2021)
Lacipirellula parvula PX69	GCA_009177095.1	(Dedysh et al., 2020)
Lautropia dentalis KCOM 2505	GCA_003892345.1	(Lim et al., 2019)
Lautropia mirabilis strain NCTC12852	GCA_900637555.1	N/A
Leifsonia shinshuensis strain INR9	GCA_014217625.1	(Benaud et al., 2021)
Leifsonia xyli subsp. cynodontis DSM 46306	GCA_000470775.1	(Monteiro-Vitorello et al., 2013)
Limnoglobus roseus strain PX52	GCA_008254045.1	(Kulichevskaya et al., 2020)
Litorilinea aerophila ATCC BAA-2444	GCA_006569185.1	(Maurais et al., 2022)
Litorilinea aerophila PRI-431	GCA_002148365.1	N/A
Longilinea arvoryzae KOME-1	GCA_001050235.2	(Matsuura et al., 2015)
Micromonospora aurantiaca ATCC 27029	GCA_000145235.1	N/A
Minicystis rosea strain DSM 24000	GCA_001931535.1	(Pal et al., 2021)
Mucilaginibacter gotjawali	GCA_002355435.1	N/A
Mucilaginibacter mallensis strain MP1X4	GCA_900105165.1	N/A
Mucilaginibacter paludis DSM 18603	GCA_000166195.3	N/A
Mycobacterium tuberculosis H37Rv	GCA_000195955.2	(Cole et al., 1998)
Mycobacterium leprae TN	GCA_000195855.1	(Cole et al., 2001)
Nakamurella endophytica CGMCC 4.7308	GCA_014646215.1	N/A
Nakamurella multipartita DSM 44233	GCA_000024365.1	(Tice et al., 2010)
Nakamurella panacisegetis strain P4-7	GCA_900104535.1	N/A
Nitratifactor salsuginis DSM 16511	GCA_000186245.1	(Anderson et al., 2011)
Nitratifactor sp.	GCA_015488535.1	(Reysenbach et al., 2020)
Novosphingobium aromaticivorans DSM 12444	GCA_000013325.1	N/A

Supplementary Table 1. (continued)

Occallatibacter savannae AB23	GCA_003131205.1	N/A
Oceanibaculum indicum USBA 36	GCA_003633955.1	N/A
Oceanibaculum pacificum MCCC 1A02656	GCA_001618175.1	N/A
Ochrobactrum anthropi PBO	GCA_015326295.1	(Gao and Sun, 2021)
Oleiharenicola lentus TWA-58	GCA_004118375.1	(Chen et al., 2020)
Opitutus sp. GAS368	GCA_900104925.1	N/A
Opitutus terrae PB90-1	GCA_000019965.1	(van Passel et al., 2011)
Pajaroellobacter abortibovis strain BTF92-0548A	GCA_001931505.1	(Welly et al., 2017)
Panacagrimonas perspica DSM 26377	GCA_004364935.1	N/A
Panacibacter ginsenosidivorans strain Gsoil1550	GCA_007971225.1	N/A
Parachlamydia acanthamoebae UV-7	GCA_000253035.1	(Collingro et al., 2011)
Parcubacteria bacterium RAAC4_OD1_1	GCA_000505065.1	(Kantor et al., 2013)
Parcubacteria bacterium SCGC AAA011-A08	GCA_000380845.1	(Rinke et al., 2013)
Pelagibacterium halotolerans strain ANSP101	GCA_013004005.1	N/A
Peristeroidobacter agariperforans strain KA5-B	GCA_004138335.1	N/A
Peristeroidobacter soli strain JW-3	GCA_004138485.1	(Huang et al., 2019)
Phenylobacterium parvum strain HYN0004	GCA_003150835.1	(Baek et al., 2019)
Phenylobacterium sp. Root1290	GCA_001425305.1	(Bai et al., 2015)
Phycisphaera mikurensis strain DSM 103959	GCA_014207395.1	N/A
Phycisphaera sp. TMED24	GCA_002168105.2	(Tully et al., 2017)
Phyllobacterium brassicacearum strain 29-15	GCA_004362695.1	N/A
Phyllobacterium trifolii strain CECT 7015	GCA_014192095.1	N/A
Phyllobacterium zundukense strain Tri-48	GCA_002764115.1	(Safronova et al., 2018)
Planctomicrobium piriforme strain DSM 26348	GCA_900113665.1	N/A
Polyangiaceae bacterium isolate DT_87	GCA_013214025.1	(Boeuf et al., 2019)
Polyangiaceae bacterium UTPRO1	GCA_002050235.1	(Lawson et al., 2017)
Polyangium fumosum strain DSM 14668	GCA_005144585.1	N/A
Polyangium aurulentum strain SDU3-1	GCA_005144635.2	N/A
Polyangium spumosum strain DSM 14734	GCA_009649845.1	N/A
Pseudorhodoplanes sinuspersici strain RIPI110	GCA_002119765.1	N/A
Pyruvatibacter mobilis strain CGMCC 1.15125	GCA_014640905.1	N/A
Rhabdothermincola sediminis SYSU G02662	GCA_014805525.1	(Liu et al., 2019)
Rhizomicrobium electricum strain DSM 21034	GCA_011762045.1	N/A
Rhizomicrobium palustre strain DSM 19867	GCA_011761565.1	N/A
Rhodanobacter denitrificans strain 2APBS1	GCA_000230695.3	N/A
Rhodanobacter glycinis strain T01E-68	GCA_008000795.1	(Lee et al., 2019a)
Rhodanobacter sp. Root480	GCA_001426645.1	(Bai et al., 2015)
Rhodomicrobium vannielii ATCC 17100	GCA_000166055.1	(Conners et al., 2021)
Rhodoplanes elegans strain DSM 11907	GCA_003258805.1	(LaSarre et al., 2018)
Rhodoplanes roseus strain DSM 5909	GCA_003258865.1	(LaSarre et al., 2018)
Rhodoplanes sp. Z2-YC6860	GCA_001579845.1	N/A
Salinispora tropica CNB-440	GCA_000016425.1	(Eustaquio et al., 2009)
Scopulibacillus darangshiensis strain DSM 19377	GCA_004341035.1	N/A
Solirubrobacter soli DSM 22325	GCA_000423665.1	N/A
Sorangium cellulosum 'So ce 56'	GCA_000067165.1	(Schneiker et al., 2007)

Supplementary Table 1. (continued)

Sorangium sp. isolate HyVt-558	GCA_011374705.1	(Zhou et al., 2020)
Spartobacteria bacterium AMD-G4	GCA_002290555.1	(Cabello-Yeves et al., 2017)
Sphingomonas melonis strain ZJ26	GCA_002504265.1	N/A
Sphingopyxis alaskensis RB2256	GCA_000013985.1	(Lauro et al., 2009)
Sphingomonas wittichii RW1	GCA_000016765.1	(Miller et al., 2010)
Steroidobacter denitrificans strain DSM 18526	GCA_001579945.1	(Yang et al., 2016)
Sulfodiicoccus acidiphilus strain JCM 31740	GCA_014648235.1	N/A
Sulfuritalea hydrogenivorans sk43H	GCA_000828635.1	(Sakai and Kurosawa, 2019)
Sulfuritalea sp. isolate Bjer_18-Q3-R1-45_MAXAC.014	GCA_016708585.1	(Singleton et al., 2021)
Syntrophobacter fumaroxidans MPOB	GCA_000014965.1	(Plugge et al., 2012)
Tepidisphaera sp. isolate RI_211	GCA_016124375.1	N/A
Terricaulis silvestris strain 0127_4	GCA_009792355.1	(Vieira et al., 2020)
Terrimonas ferruginea DSM 30193	GCA_000425585.1	N/A
Terrimonas sp. strain NS-102	GCA_003075435.1	N/A
Thermocladium modestius strain JCM 10088	GCA_014646535.1	N/A
Thermogutta terrifontis strain R1	GCA_002277955.1	(Elcheninov et al., 2017)
Variibacter gotjawalensis GJW-30	GCA_002355335.1	(Lee et al., 2016)
Variovorax ginsengisoli strain S09.D	GCA_006438845.1	(Altshuler et al., 2019)
Variovorax soli NBRC 106424	GCA_001591385.1	N/A
Verrucomicrobium spinosum DSM 4136	GCA_000172155.1	(Xing et al., 2019)
Zavarzinella formosa DSM 19928	GCA_000255705.1	(Guo et al., 2012)
Zymomonas mobilis subsp. mobilis ZM4	GCA_000007105.1	(Seo et al., 2005)

Supplementary Table 1. (continued)

Availability of data and materials

The datasets supporting the conclusions of this article are publicly available in the National Center for Biotechnology Information (NCBI) repository under BioProject PRJNA1124010.

Acknowledgements

We would like to thank David Pressler and the Botanical Garden of the University of Vienna for maintaining the plant in cultivation, and the Tyrol County administration for issuing necessary permits.

Additional information

Authors' contribution

GAV, CZ, DFS and OV conceptualized and designed the study. GAV and LZ generated the datasets. GAV and JC performed the data analysis. GAV and JC generated the figures and drafted the manuscript. ARMA and RLM revised and contributed to the manuscript. All authors read and approved the final manuscript.

Funding

This study was supported by a grant from the Austrian Science Fund (FWF): P29556 given to RM, and the PhD program TU Wien bioactive.

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Abbreviations

ANI: average nucleotide identity
ASV: amplicon sequence variant
BGC: biosynthetic gene cluster
MAG: metagenome assembled genome
MIMAG: Minimum Information about a Metagenome-Assembled Genome
OTU: operational taxonomic unit
PERMANOVA: permutational multivariate analysis of variance
PGPR: plant growth promoting Rhizobacteria
RAB: root associated bacteria
SM: secondary metabolite

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Additional files

Supplementary Table 2

Table 6

Table 7

Table 8

Table 9

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Peer reviews

Reviewer #1 (Public review):

Summary:

The microbiota of Dactylorhiza traunsteineri, an endangered marsh orchid, forms complex root associations that support plant health. Using 16S rRNA sequencing, we identified dominant bacterial phyla in its rhizosphere, including Proteobacteria, Actinobacteria, and Bacteroidota. Deep shotgun metagenomics revealed high-quality MAGs with rich metabolic and biosynthetic potential. This study provides key insights into root-associated bacteria and highlights the rhizosphere as a promising source of bioactive compounds, supporting both microbial ecology research and orchid conservation.

Strengths:

The manuscript presents an investigation of the bacterial communities in the rhizosphere of D. traunsteineri using advanced metagenomic approaches. The topic is relevant, and the techniques are up-to-date; however, the study has several critical weaknesses.

Weaknesses:

(1) Title: The current title is misleading. Given that fungi are the primary symbionts in orchids and were not analyzed in this study (nor were they included among other microbial groups), the use of the term "microbiome" is not appropriate. I recommend replacing it with "bacteriome" to better reflect the scope of the work.

(2) Line 124: The phrase "D. traunsteineri individuals were isolated" seems misleading. A more accurate description would be "individuals were collected", as also mentioned in line 128.

(3) Experimental design: The major limitation of this study lies in its experimental design. The number of plant individuals and soil samples analyzed is unclear, making it difficult to assess the statistical robustness of the findings. It is also not well explained why the orchids were collected two years before the rhizosphere soil samples. Was the rhizosphere soil collected from the same site and from remnants of the previously sampled individuals in 2018? This temporal gap raises serious concerns about the validity of the biological associations being inferred.

(4) Low sample size: In lines 249-251 (Results section), the authors mention that only one plant individual was used for identifying rhizosphere bacteria. This is insufficient to produce scientifically robust or generalizable conclusions.

(5) Contextual limitations: Numerous studies have shown that plant-microbe interactions are influenced by external biotic and abiotic factors, as well as by plant age and population

structure. These elements are not discussed or controlled for in the manuscript. Furthermore, the ecological and environmental conditions of the site where the plants and soil were collected are poorly described. The number of biological and technical replicates is also not clearly stated.

(6) Terminology: Throughout the manuscript, the authors refer to the "microbiome," though only bacterial communities were analyzed. This terminology is inaccurate and should be corrected consistently.

Considering the issues addressed, particularly regarding experimental design and data interpretation, significant improvements to the study are needed.

https://doi.org/10.7554/eLife.108126.1.sa2

Reviewer #2 (Public review):

Summary:

The authors aim to provide an overview of the D. traunsteineri rhizosphere microbiome on a taxonomic and functional level, through 16S rRNA amplicon analysis and shotgun metagenome analysis. The amplicon sequencing shows that the major phyla present in the microbiome belong to phyla with members previously found to be enriched in rhizospheres and bulk soils. Their shotgun metagenome analysis focused on producing metagenome assembled genomes (MAGs), of which one satisfies the MIMAG quality criteria for high-quality MAGs and three those for medium-quality MAGs. These MAGs were subjected to functional annotations focusing on metabolic pathway enrichment and secondary metabolic pathway biosynthetic gene cluster analysis. They find 1741 BGCs of various categories in the MAGs that were analyzed, with the high-quality MAG being claimed to contain 181 SM BGCs. The authors provide a useful, albeit superficial, overview of the taxonomic composition of the microbiome, and their dataset can be used for further analysis.

The conclusions of this paper are not well-supported by the data, as the paper only superficially discusses the results, and the functional interpretation based on taxonomic evidence or generic functional annotations does not allow drawing any conclusions on the functional roles of the orchid microbiota.

Weaknesses:

The authors only used one individual plant to take samples. This makes it hard to generalize about the natural orchid microbiome.

The authors use both 16S amplicon sequencing and shotgun metagenomics to analyse the microbiome. However, the authors barely discuss the similarities and differences between the results of these two methods, even though comparing these results may be able to provide further insights into the conclusions of the authors. For example, the relative abundance of the ASVs from the amplicon analysis is not linked to the relative abundances of the MAGs.

Furthermore, the authors discuss that phyla present in the orchid microbiome are also found in other microbiomes and are linked to important ecological functions. However, their results reach further than the phylum level, and a discussion of genera or even species is lacking. The phyla that were found have very large within-phylum functional variability, and reliable functional conclusions cannot be drawn based on taxonomic assignment at this level, or even the genus level (Yan et al. 2017).

Additionally, although the authors mention their techniques used, their method section is sometimes not clear about how samples or replicates were defined. There are also inconsistencies between the methods and the results section, for example, regarding the prediction of secondary metabolite biosynthetic gene clusters (BGCs).

The BGC prediction was done with several tools, and the unusually high number of found BGCs (181 in their high-quality MAG) is likely due to false positives or fragmented BGCs. The numbers are much higher than any numbers ever reported in literature supported by functional evidence (Amos et al, 2017), even in a prolific genus like Streptomyces (Belknap et al., 2020). This caveat is not discussed by the authors.

The authors have generated one high-quality MAG and three medium-quality MAGs. In the discussion, they present all four of these as high-quality, which could be misleading. The authors discuss what was found in the literature about the role of the bacterial genera/phyla linked to these MAGs in plant rhizospheres, but they do not sufficiently link their own analysis results (metabolic pathway enrichment and biosynthetic gene cluster prediction) to this discussion. The results of these analyses are only presented in tables without further explanation in either the results section or the discussion, even though there may be interesting findings. For example, the authors only discuss the class of the BGCs that were found, but don't search for experimentally verified homologs in databases, which could shed more light on the possible functional roles of BGCs in this microbiome.

In the conclusions, the authors state: "These analyses uncovered potential metabolic capabilities and biosynthetic potentials that are integral to the rhizosphere's ecological dynamics." I don't see any support for this. Mentioning that certain classes of BGCs are present is not enough to make this claim, in my opinion. Any BGC is likely important for the ecological niche the bacteria live in. The fact that rhizosphere bacteria harbour BGCs is not surprising, and it doesn't tell us more than is already known.

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Amos GCA, Awakawa T, Tuttle RN, Letzel AC, Kim MC, Kudo Y, Fenical W, Moore BS, Jensen PR. Comparative transcriptomics as a guide to natural product discovery and biosynthetic gene cluster functionality. Proc Natl Acad Sci U S A. 2017 Dec 26;114(52):E11121-E11130.

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https://doi.org/10.7554/eLife.108126.1.sa1

Author response:

> ***Reviewer #1 (Public review):***
>
> *The microbiota of Dactylorhiza traunsteineri, an endangered marsh orchid, forms complex root associations that support plant health. Using 16S rRNA sequencing, we identified dominant bacterial phyla in its rhizosphere, including Proteobacteria,*

> *Actinobacteria, and Bacteroidota. Deep shotgun metagenomics revealed high-quality MAGs with rich metabolic and biosynthetic potential. This study provides key insights into root-associated bacteria and highlights the rhizosphere as a promising source of bioactive compounds, supporting both microbial ecology research and orchid conservation.*
>
> *The manuscript presents an investigation of the bacterial communities in the rhizosphere of D. traunsteineri using advanced metagenomic approaches. The topic is relevant, and the techniques are up-to-date; however, the study has several critical weaknesses.*

We thank the reviewer for their careful reading of our manuscript and for the constructive comments. We will revise the manuscript substantially. Our responses to the specific points are below:

> *(1) Title: The current title is misleading. Given that fungi are the primary symbionts in orchids and were not analyzed in this study (nor were they included among other microbial groups), the use of the term "microbiome" is not appropriate. I recommend replacing it with "bacteriome" to better reflect the scope of the work.*

In the revised manuscript, we will expand the Results (shotgun sequencing) and Discussion to also include fungal taxa. With these additions, the use of the term microbiome will accurately reflect the inclusion of both bacterial and fungal components.

> *(2) Line 124: The phrase "D. traunsteineri individuals were isolated" seems misleading. A more accurate description would be "individuals were collected", as also mentioned in line 128.*

This ambiguity will be corrected in the revised manuscript.

> *(3) Experimental design: The major limitation of this study lies in its experimental design. The number of plant individuals and soil samples analyzed is unclear, making it difficult to assess the statistical robustness of the findings. It is also not well explained why the orchids were collected two years before the rhizosphere soil samples. Was the rhizosphere soil collected from the same site and from remnants of the previously sampled individuals in 2018? This temporal gap raises serious concerns about the validity of the biological associations being inferred.*

In the revised manuscript, we will explicitly state the number of individuals and soil samples included in the study, and we will more clearly describe the sequence of sampling events. We will also add a dedicated statement in the Discussion addressing the temporal gap between plant sampling and rhizosphere soil collection, acknowledging that this is a limitation of the study.

> *(4) Low sample size: In lines 249-251 (Results section), the authors mention that only one plant individual was used for identifying rhizosphere bacteria. This is insufficient to produce scientifically robust or generalizable conclusions.*

In the revised manuscript, we will clearly state that only one rhizosphere sample was available and will frame the study as exploratory in nature. We will explicitly acknowledge this limitation in both the Methods and Discussion, and we will temper our conclusions accordingly.

> *(5) Contextual limitations: Numerous studies have shown that plant-microbe interactions are influenced by external biotic and abiotic factors, as well as by plant age and population structure. These elements are not discussed or controlled for in the manuscript. Furthermore, the ecological and environmental conditions of the site where*

> *the plants and soil were collected are poorly described. The number of biological and technical replicates is also not clearly stated.*

In the revised manuscript, we will expand the description of the collection site and environmental conditions to the extent supported by our records. We will also clearly state the number of biological and technical replicates used for each analysis. In the Discussion, we will explicitly acknowledge that plant age, environmental variables, and other biotic/abiotic factors may influence plant–microbe interactions and were not directly assessed in this study.

> *(6) Terminology: Throughout the manuscript, the authors refer to the "microbiome," though only bacterial communities were analyzed. This terminology is inaccurate and should be corrected consistently.*

As noted in our response to point (1), we will revise terminology throughout the manuscript to ensure consistency and to accurately reflect the expanded bacterial and fungal coverage in the revised version.

> ***Reviewer #2 (Public review):***
>
> *The authors aim to provide an overview of the D. traunsteineri rhizosphere microbiome on a taxonomic and functional level, through 16S rRNA amplicon analysis and shotgun metagenome analysis. The amplicon sequencing shows that the major phyla present in the microbiome belong to phyla with members previously found to be enriched in rhizospheres and bulk soils. Their shotgun metagenome analysis focused on producing metagenome assembled genomes (MAGs), of which one satisfies the MIMAG quality criteria for high-quality MAGs and three those for medium-quality MAGs. These MAGs were subjected to functional annotations focusing on metabolic pathway enrichment and secondary metabolic pathway biosynthetic gene cluster analysis. They find 1741 BGCs of various categories in the MAGs that were analyzed, with the high-quality MAG being claimed to contain 181 SM BGCs. The authors provide a useful, albeit superficial, overview of the taxonomic composition of the microbiome, and their dataset can be used for further analysis.*
>
> *The conclusions of this paper are not well-supported by the data, as the paper only superficially discusses the results, and the functional interpretation based on taxonomic evidence or generic functional annotations does not allow drawing any conclusions on the functional roles of the orchid microbiota.*

We thank the reviewer for their thoughtful and constructive assessment of our manuscript. The comments have been very helpful in identifying areas where the clarity, structure, and interpretation of our work can be improved. Our responses to the specific points are below:

> *(1) The authors only used one individual plant to take samples. This makes it hard to generalize about the natural orchid microbiome.*

We agree with the reviewer that the limited number of plant individuals restricts the generality of the conclusions. In the revised manuscript, we will clearly state that only one rhizosphere sample was available for analysis and will frame the study as exploratory. We will also explicitly acknowledge this limitation in the Discussion and ensure that our interpretations and conclusions remain appropriately cautious.

> *(2) The authors use both 16S amplicon sequencing and shotgun metagenomics to analyse the microbiome. However, the authors barely discuss the similarities and differences between the results of these two methods, even though comparing these results may be able to provide further insights into the conclusions of the authors. For example, the relative abundance of the ASVs from the amplicon analysis is not linked to the relative abundances of the MAGs.*

In the revised manuscript, we will expand the Results and Discussion to include a clearer comparison between the taxonomic profiles derived from 16S amplicon sequencing and those obtained from shotgun metagenomic binning.

> *(3) Furthermore, the authors discuss that phyla present in the orchid microbiome are also found in other microbiomes and are linked to important ecological functions. However, their results reach further than the phylum level, and a discussion of genera or even species is lacking. The phyla that were found have very large within-phylum functional variability, and reliable functional conclusions cannot be drawn based on taxonomic assignment at this level, or even the genus level (Yan et al. 2017).*

In the revised manuscript, we will incorporate taxonomic discussion at finer resolution where reliable assignments are available. We will also revise the Discussion to avoid overinterpreting phylum-level taxonomy in terms of ecological function.

> *(4) Additionally, although the authors mention their techniques used, their method section is sometimes not clear about how samples or replicates were defined. There are also inconsistencies between the methods and the results section, for example, regarding the prediction of secondary metabolite biosynthetic gene clusters (BGCs).*

In the revised Methods section, we will clearly define the number and type of samples included in each analysis, specify the number of replicates and how they were handled, and provide a clearer description of the biosynthetic gene cluster (BGC) prediction workflow, including the tools used and how results were interpreted.

> *(5) The BGC prediction was done with several tools, and the unusually high number of found BGCs (181 in their high-quality MAG) is likely due to false positives or fragmented BGCs. The numbers are much higher than any numbers ever reported in literature supported by functional evidence (Amos et al, 2017), even in a prolific genus like Streptomyces (Belknap et al., 2020). This caveat is not discussed by the authors.*

We thank the reviewer for this important point. Our original intention was to present the BGC predictions as a resource for future exploration, which is why multiple tools were used. However, we understand how this approach may lead to confusion, particularly regarding the confidence level of the predicted clusters and the potential inflation of counts due to assembly fragmentation or tool sensitivity. In the revised manuscript, we will thoroughly revise this section to clearly distinguish highconfidence predictions from more exploratory findings. We will focus on results supported by stronger evidence, explicitly qualify lower-confidence predictions as putative, and temper any functional interpretations accordingly.

> *(6) The authors have generated one high-quality MAG and three medium-quality MAGs. In the discussion, they present all four of these as high-quality, which could be misleading. The authors discuss what was found in the literature about the role of the bacterial genera/phyla linked to these MAGs in plant rhizospheres, but they do not sufficiently link their own analysis results (metabolic pathway enrichment and biosynthetic gene cluster prediction) to this discussion. The results of these analyses are only presented in tables without further explanation in either the results section or the discussion, even though there may be interesting findings. For example, the authors only discuss the class of the BGCs that were found, but don't search for experimentally verified homologs in databases, which could shed more light on the possible functional roles of BGCs in this microbiome.*

In the revised manuscript, we will ensure that MAG quality is described accurately and consistently throughout, distinguishing clearly between high-quality and medium-quality bins according to accepted standards.

> *(7) In the conclusions, the authors state: "These analyses uncovered potential metabolic capabilities and biosynthetic potentials that are integral to the rhizosphere's ecological dynamics." I don't see any support for this. Mentioning that certain classes of BGCs are present is not enough to make this claim, in my opinion. Any BGC is likely important for the ecological niche the bacteria live in. The fact that rhizosphere bacteria harbour BGCs is not surprising, and it doesn't tell us more than is already known.*

In the revised manuscript, we will rewrite the conclusion to reflect a more cautious interpretation, focusing on the potential metabolic and biosynthetic capabilities suggested by the data without asserting ecological roles that cannot be directly supported. These capabilities will be presented as hypotheses for future investigation rather than established ecological features.

https://doi.org/10.7554/eLife.108126.1.sa0